Complete Bounded Repair Ports:
Transfer, Reliability, and Geometric Structure

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Abstract

For a linear code and a distinguished coordinate, the complete bounded repair port records every dual-support recovery using at most a prescribed number of helpers. We keep three layers of this object visible: its support clutter, its scalar recovery coefficients, and its reliability under helper failures.

We prove an exact weighted-functional transfer theorem for concatenation and identify the persistent pointed obstruction to replicating a prescribed port. Every fixed represented port satisfying this obstruction bound occurs with positive density in an asymptotically good fixed-alphabet family. The same object has an exact deletion–contraction reliability calculus and a radius-truncated BEC EXIT hierarchy whose successive differences give the cheapest available repair radius. At full radius, repair reliability is a specialization of the Las Vergnas polynomial of the elementary perspective $M\backslash x \to M/x$; the bounded-radius filtration is extra information.

Two characteristic-three families illustrate the distinction. A twisted-cubic–axis code has exact coordinatewise matching and transversal rows and transfers strictly beyond the ordinary support-distance gate. A quartic normal rational curve with its nucleus has harmonic-quadruple circuits, a Steiner $S(3, 4, q + 1)$ nucleus port, and parallel repairs at the nucleus, whereas every curve-target repair has the nucleus as a compulsory helper. These parallel and series geometries separate bounded-radius from full-port information.

1 Introduction

Locality asks how many surviving symbols suffice to recover one unavailable coordinate. Availability asks for many disjoint choices of helpers. Both are standard in locally repairable codes [7, 14, 10], but neither describes the whole local failure event: two codes can have the same locality and matching number while their permitted repair sets overlap very differently.

The natural pointed object is the family of *all* bounded dual-support repairs. We call it a complete bounded repair port. Its matching number is disjoint availability, its transversal number minus one is worst-case helper failure tolerance, and its monotone Boolean function gives exact stochastic reliability. The coefficient layer matters as well: a repair support comes from a scalar equation, and concatenation is controlled by the minimum cost of realizing the corresponding block functional rather than by support alone.

For a target x, let $\mathcal{P}_x^{(\le r)}(C)$ be the dual words normalized by $w_x = 1$ and supported on x and at most r helpers. This coefficient port records the scalar recovery equations, not only their supports.

Theorem 1.1 (MDS coefficient-port reconstruction)**.** *Let $C \le \mathbb{F}_q^E$ be an $[n, k]_q$ MDS code with $1 \le k < n$, and let $x \in E$. Then*

$$\operatorname{span} \mathcal{P}_x^{(\le k)}(C) = C^{\perp}, \qquad \rho_x(C) = k,$$

where $\rho_x(C)$ is the least reconstruction radius. The support projection of the minimum coefficient port is the complete k-uniform clutter on $E \setminus \{x\}$.

The support clutter in this theorem is generic: it is the same for every MDS code with these parameters. The normalized coefficients nevertheless span the represented dual code. The proof chooses minimum dual words with a common core of $k-1$ helpers and distinct private coordinates, producing a basis of $C^{\perp}$. Thus the coefficient layer retains represented information that support-only locality discards.

Figure 1 gives the proof spine in one view. The exact transfer theorem concerns the first two port layers; reliability is then determined by the transported support layer.

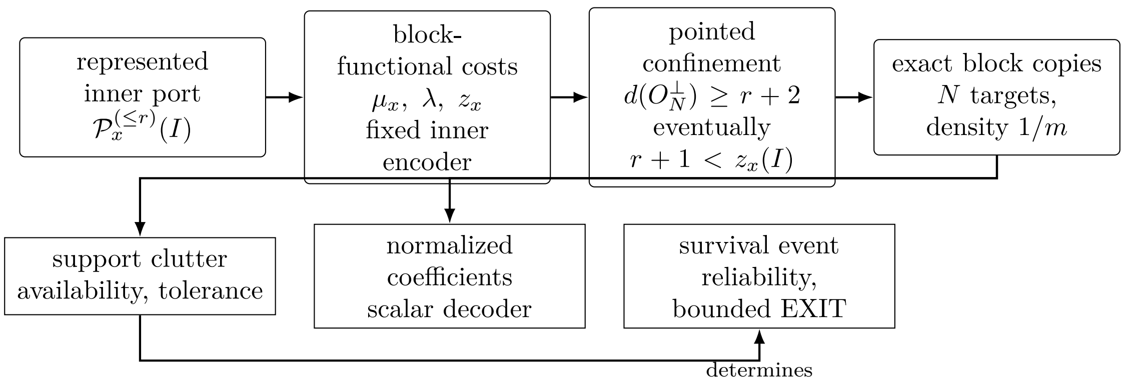

Figure 1: The transfer spine. In the fixed-inner, L-linear concatenation regime, weighted block-functional costs give an exact confinement gate. When $d(O_N^{\perp}) \to \infty$ and $r+1 < z_x(I)$, support and normalized coefficient ports are copied into every block. Reliability is derived from the copied support clutter; the diagram makes no necessity claim for arbitrary asymptotically good families.

The paper has six parts. Section 2 fixes the complete pointed object and separates support, coefficient, and probability layers. Section 3 gives exact and usable concatenation gates. Section 4 turns the pointed gate into a prescribed positive-density realization theorem. Section 5 develops reliability and bounded EXIT. Section 6 identifies the full-port polynomial as standard pointed-Tutte structure and isolates what the radius filtration adds. Section 7 compares cubic–axis and quartic–nucleus flagships.

The asymptotic coding ingredients, reliability identities, and pointed-Tutte polynomial are classical. We develop their exact organization around the complete bounded pointed object, together with the transfer boundary, prescribed-port consequence, and the two exact geometric port inventories. We make no minimum-bandwidth claim, no full-MAP claim for a radius cutoff, and no threshold claim for harmonic closure. In the literature reviewed, we did not locate the exact complete-port confinement criterion or the two coordinatewise port inventories [14, 11, 8, 5]; this is a bounded comparison, not a priority claim.

2 Complete bounded repair ports

Let $C \le \mathbb{F}_q^X$ be a linear code, let $x \in X$, and let $r \ge 0$.

Throughout, whenever matching, transversal, failure tolerance $\tau - 1$, or a minimum-blocker leading term is asserted, the minimal repair clutter is nonempty and does not contain the empty helper set. The support, coefficient, and reliability definitions themselves remain valid in the two degenerate cases: an empty port never repairs, while an empty repair edge repairs without a helper. We state those cases separately rather than assign them an adversarial-tolerance interpretation.

Definition 2.1 (Support layer)**.** The complete radius-r repair hypergraph at x is

$$\mathcal{H}_x^{(\le r)}(C) = \Big\{ R \subseteq X \setminus \{x\} : |R| \le r,\ \exists w \in C^{\perp},\ w_x \neq 0,\ \operatorname{supp}(w) \setminus \{x\} = R \Big\}.$$

Its minimal members form the repair clutter $\min \mathcal{H}_x^{(\le r)}(C)$. Write $\nu_x^{(\le r)}$ and $\tau_x^{(\le r)}$ for its matching and transversal numbers.

The distinction between the complete hypergraph and its minimal clutter is useful: the former records every bounded witness, while every monotone support invariant below depends only on the latter.

Proposition 2.2 (Basic invariants)**.** *Assume the empty helper set is not a repair. Then*

$$\nu_x^{(\le r)} \le \tau_x^{(\le r)}.$$

Both numbers are unchanged after deleting nonminimal repair sets, so either the complete hypergraph or its minimal clutter may be used for these optima.

Proof. A matching of m nonempty edges forces every transversal to select at least one point from each edge. Minimalization preserves the upward closure of the success event, hence preserves matching and transversal optima. □

Definition 2.3 (Coefficient layer)**.** Write

$$\mathcal{P}_x^{(\le r)}(C) = \{ w \in C^{\perp} : w_x = 1,\ |\operatorname{supp}(w) \setminus \{x\}| \le r \}$$

for the normalized coefficient port. For $R \in \mathcal{H}_x^{(\le r)}(C)$, its coefficient fiber is the set of normalized coefficient vectors

$$\left(-\frac{w_y}{w_x} \right)_{y \in R} \qquad (w \in C^{\perp},\ \operatorname{supp}(w) = R \cup \{x\}).$$

Each vector gives the exact recovery equation $c_x = \sum_{y \in R} (-w_y/w_x) c_y$. Operationally, a helper transmits its stored scalar in $\mathbb{F}_q$ and the decoder applies the displayed coefficient. Thus a radius-r repair contacts at most r helpers and downloads one whole base-field symbol from each. This scalar exact-repair model does not optimize subsymbol access or repair bandwidth in an array-code model [9, 22, 16].

Definition 2.4 (Reconstruction radius)**.** The coefficient port *reconstructs* C at radius r when its linear span is $C^{\perp}$; equivalently, double duality recovers C from the port. Its reconstruction radius is the least such r, or infinity if no such radius exists.

Proof of Theorem 1.1. The dual has dimension $n-k$ and minimum distance $k+1$. Given any $(k+1)$-set $T \subseteq E$, restrict $C^{\perp}$ to $E \setminus T$. Its domain has dimension $n-k$ and its codomain dimension $n-k-1$, so the restriction has a nonzero kernel vector y supported on T. The dual-distance bound forces $\operatorname{supp}(y) = T$. If $x \in T$, then $y_x \neq 0$ and scaling gives a normalized repair word. Thus every k-subset of $E \setminus \{x\}$ is a minimum helper set. Conversely, any normalized repair with at most k helpers has total weight at most $k+1$, so the same distance bound forces equality. This proves the support-clutter assertion.

Choose $Q \subseteq E \setminus \{x\}$ with $|Q| = k-1$ and set $U = E \setminus (\{x\} \cup Q)$. For every $t \in U$, choose the normalized minimum dual word y_t supported on $\{x,t\} \cup Q$. The coordinate t is nonzero in y_t and zero in every y_s with $s \neq t$. Evaluating a linear relation at the private coordinates t shows that the y_t are independent. There are $|U| = n-k = \dim C^\perp$ of them, so they form a basis of $C^\perp$ contained in the radius-k coefficient port. Double duality therefore recovers C. No smaller radius contains any normalized word, since that would produce a nonzero dual word of weight at most k. Hence $\rho_x(C) = k$. □

Definition 2.5 (Probability layer)**.** Let $A \subseteq X \setminus \{x\}$ be the surviving helpers and put

$$f_x^{(\leq r)}(A) = 1 \quad \iff \quad \exists R \in \mathcal{H}_x^{(\leq r)}(C) \text{ with } R \subseteq A.$$

For independent survival probabilities (s_y), the reliability is $R_x^{(\leq r)}(s) = \mathbb{E}[f_x^{(\leq r)}]$.

Exact equality of complete ports preserves all support-derived invariants: matching, transversals, blocker clutters, multivariate reliability, and every correlated-law reliability after transporting the helper labels. Coefficient fibers require the finer functional data used next.

3 Exact weighted-functional transfer

Let $I \leq \mathbb{F}_q^m$ be an inner $[m,k,d]_q$ code and identify its message space with $L = \mathbb{F}_{q^k}$ through an $\mathbb{F}_q$-linear encoder $e : L \to I$. For $w \in \mathbb{F}_q^m$, define the induced block functional

$$\Phi_I(w) : L \longrightarrow \mathbb{F}_q, \qquad a \longmapsto \langle w, e(a) \rangle.$$

The zero fiber is $I^\perp$. For $\beta \in \mathrm{Hom}_{\mathbb{F}_q}(L, \mathbb{F}_q)$ put

$$\lambda(\beta) = \min_{\Phi_I(w)=\beta} \mathrm{wt}(w), \qquad \mu_x(\beta) = \min_{\substack{\Phi_I(w)=\beta \\ w_x \neq 0}} \mathrm{wt}(w),$$

with value ∞ for an empty constrained fiber.

Let $O \leq L^J$ be an outer code. Its functional dual consists of tuples $(\beta_j)_{j \in J}$ satisfying

$$\sum_{j \in J} \beta_j(a_j) = 0 \qquad \text{for every } a = (a_j) \in O.$$

The weighted cost of such a tuple is $\sum_j \lambda(\beta_j)$. This is the ordinary quotient/coset-leader weight attached to the inner encoder; the pointed use below is the repair-specific part. Write

$$\mathrm{fwt}(\beta) = \big|\{j : \beta_j \neq 0\}\big|.$$

Thus the functional-dual tuples have three disjoint strata: $\mathrm{fwt}(\beta) = 0$, $\mathrm{fwt}(\beta) = 1$, and $\mathrm{fwt}(\beta) \geq 2$. Keeping the singleton stratum is essential when an outer coordinate projection is not surjective. If every coordinate projection of O is surjective, a functional-dual tuple cannot have functional weight one.

The all-zero functional sector has the fixed pointed cost

$$z_x(I) = \begin{cases} \mu_x(0) + d(I^\perp), & I^\perp \neq 0, \\ \infty, & I^\perp = 0. \end{cases}$$

Theorem 3.1 (Exact pointed confinement and transfer)**.** *Assume $|J| \geq 2$, and fix an outer block j and an inner coordinate x. The minimum weight of a concatenated dual witness through (j,x) that is not an embedded inner-dual witness is*

$$\min\left\{ z_x(I), \min_{\substack{(\beta_\ell) \text{ in the functional dual of } O \\ (\beta_\ell) \neq 0}} \left(\mu_x(\beta_j) + \sum_{\ell \neq j} \lambda(\beta_\ell) \right) \right\},$$

where a minimum over an empty set is ∞. Consequently, if this pointed cost is greater than $r+1$, then

$$\mathcal{H}^{(\leq r)}_{(j,x)}(O \circ I) = \mathrm{embed}_j\big(\mathcal{H}^{(\leq r)}_x(I)\big).$$

Under the same hypothesis, zero-extension is a bijection

$$\mathcal{P}^{(\leq r)}_x(I) \xrightarrow{\sim} \mathcal{P}^{(\leq r)}_{(j,x)}(O \circ I).$$

In particular it suffices that

$$r + 1 < 2d(I^\perp) \quad \text{and} \quad \sum_j \lambda(\beta_j) \geq r + 2$$

for every nonzero functional-dual tuple.

Proof. Let $w = (w_\ell)_{\ell \in J}$ be a block word and set $\beta_\ell = \Phi_I(w_\ell)$. For every outer word $a = (a_\ell) \in O$,

$$\langle w, e(a) \rangle = \sum_{\ell \in J} \langle w_\ell, e(a_\ell) \rangle = \sum_{\ell \in J} \beta_\ell(a_\ell).$$

Hence $w \in (O \circ I)^\perp$ if and only if $\beta = (\beta_\ell)$ belongs to the functional dual of O. This calculation is the only concatenated-dual decomposition used below; no fiber enumerator is needed.

Fix such a tuple β. The constraints on the blocks are independent. At the target, the condition $w_j(x) \neq 0$ gives minimum $\mu_x(\beta_j)$. At every other block the minimum is $\lambda(\beta_\ell)$. Each ordinary fiber is nonempty because a functional on the encoded inner space extends to the ambient coordinate space, and its minimum is attained because Hamming weight takes values in the well-ordered set $\mathbb{N}$. The constrained target fiber is either empty, in which case its cost and the fixed-fiber cost are infinite, or has an attained minimum by the same argument. Therefore the exact pointed minimum in the fixed fiber β is

$$\mu_x(\beta_j) + \sum_{\ell \neq j} \lambda(\beta_\ell).$$

If $\beta \neq 0$, a realization cannot be an embedded inner-dual word in block j, since every embedded inner-dual word induces the zero functional in every block. Taking the minimum over the nonzero functional dual gives the second term in the theorem. This minimum includes both remaining strata: functional weight one and functional weight at least two.

It remains to analyze $\beta = 0$. Then every w_ℓ lies in $I^\perp$. A pointed realization that is not embedded in block j consists of a word in $I^\perp$ that is nonzero at x, together with a nonzero word of $I^\perp$ in at least one off-target block. Conversely, placing any such pair in two blocks and putting zero elsewhere gives a pointed nonembedded concatenated-dual word. Since $|J| \geq 2$, the two choices can be made independently, and their exact minimum is $\mu_x(0) + d(I^\perp) = z_x(I)$. If either required word does not exist, both the constrained minimum and $z_x(I)$ are infinite. The zero and nonzero functional sectors partition all pointed nonembedded witnesses, proving the displayed minimum formula.

Suppose that this minimum is greater than $r+1$, and take a radius-r pointed repair witness in the concatenated code. Its total weight is at most $r+1$, so it cannot be nonembedded. It is therefore the zero-extension of an inner-dual word in block j. Deleting the target coordinate identifies its helper support with the image under embed_j of a radius-r inner helper support. This proves one inclusion of the asserted support-port equality. If the witness is normalized at the target, its nonzero block is the correspondingly normalized inner word, so this identification also gives the asserted coefficient-port bijection. Conversely, the zero-extension of every radius-r inner repair word is orthogonal to the concatenated code, and its helper support is the corresponding embedded support; normalization is unchanged. This proves the reverse inclusions in both layers.

Finally, whenever the zero-sector cost is finite, $\mu_x(0) \geq d(I^\perp)$, so $z_x(I) \geq 2d(I^\perp)$. For a nonzero functional tuple, $\mu_x(\beta_j) \geq \lambda(\beta_j)$, and hence its pointed cost is at least $\sum_\ell \lambda(\beta_\ell)$. The two displayed numerical hypotheses therefore put every pointed nonembedded witness above weight $r+1$, and the transfer conclusion follows. □

The functional-dual decomposition is classical concatenation machinery [2, Theorem 2.3][3, Theorem 2.1]. The weighted statement matters because ordinary functional support distance can be strictly weaker.

Corollary 3.2 (Strict weighted transfer)**.** *Let I be the completed $[20,4,9]_9$ cubic–axis code of Section 7, identify its message space with $L = \mathbb{F}_{9^4}$, and let $a : L \to L$ be an $\mathbb{F}_9$-linear Singer automorphism chosen as in the proof below. The generalized single-parity-check outer code*

$$O_a = \{u \in L^5 : u_0 + a(u_1) + u_2 + u_3 + u_4 = 0\}$$

has functional-dual support distance five, but its nonzero weighted realization cost for I is at least six. Thus the ordinary support-distance condition at radius four fails while the exact weighted gate holds, and every radius-four support and coefficient port transfers to each of the five blocks of $O_a \circ I$.

Proof. The nonzero projective functionals on the four-dimensional $\mathbb{F}_9$-space L form $(9^4-1)/(9-1) = 820$ classes. The twenty coordinate functionals of I give twenty distinct classes because $d(I^\perp) = 3$. Singer's regular action on these projective classes [17] supplies a multiplier a whose translate of this twenty-set is disjoint from the original set: at most $20^2 = 400 < 820$ group elements can map some selected class to another selected class.

Every functional-dual tuple of O_a is $(f, f \circ a, f, f, f)$. If $f \neq 0$, all five entries are nonzero, so the functional support distance is exactly five. A weight-five realization would have unit cost in every block. In particular, both f and $f \circ a$ would lie in the two disjoint unit-cost class sets, a contradiction. Hence every nonzero tuple costs at least six. The zero sector also costs at least $2d(I^\perp) = 6$, so Theorem 3.1 with $r = 4$ gives the two exact port equalities. The displayed parity check makes every outer coordinate projection surjective. Taking any nonzero f shows at the same time that functional distance six is false. □

4 Prescribed positive-density realization

The pointed zero-functional sector persists even when outer dual distance grows.

Theorem 4.1 (Prescribed represented ports)**.** *Let $O_N \leq L^N$ be L-linear outer codes with $d(O_N^\perp) \to \infty$, and let $C_N = O_N \circ I$. For all sufficiently large N, every pointed dual witness through (j, x) of weight at most $r+1$ is confined to block j if and only if*

$$r + 1 < z_x(I).$$

Under this condition the radius-r port at x occurs at the N targets (j,x), hence with density $1/m$ in C_N. More precisely, at every block

$$\mathcal{H}^{(\le r)}_{(j,x)}(C_N)=\mathrm{embed}_j\big(\mathcal{H}^{(\le r)}_x(I)\big),\qquad \mathcal{P}^{(\le r)}_{(j,x)}(C_N)=\mathrm{embed}_j\big(\mathcal{P}^{(\le r)}_x(I)\big).$$

Proof. Every $\mathbb{F}_q$-linear functional on L has a unique form $a\mapsto \mathrm{Tr}_{L/\mathbb{F}_q}(ba)$, by nondegeneracy of the trace pairing. Let $\beta=(\beta_j)$ lie in the functional dual of the restricted outer code, and write $\beta_j(a)=\mathrm{Tr}(b_ja)$. Since O_N is L-linear, applying β to every scalar multiple ca of every $a\in O_N$ shows

$$\mathrm{Tr}\left(c\sum_j b_ja_j\right)=0\quad\text{for every } c\in L.$$

Trace nondegeneracy gives $(b_j)\in O_N^{\perp}$. Conversely, every outer dual word gives such a functional tuple. The trace representation is coordinatewise injective, so it preserves support exactly. Thus the functional-dual support distance is $d(O_N^{\perp})$.

Fix r. For all sufficiently large N, $d(O_N^{\perp})\ge r+2$. Every realization of a nonzero functional tuple then has weight at least $r+2$: each nonzero functional requires a nonzero inner block. The exact pointed decomposition of Theorem 3.1 consequently leaves only the zero-functional sector in the bounded range. For $N\ge 2$, its exact cost is the block-independent quantity $z_x(I)$. Hence confinement through every (j,x) holds exactly when $r+1<z_x(I)$.

If $r+1<z_x(I)$, Theorem 3.1 identifies both support and normalized coefficient ports by zero-extension. If $r+1\ge z_x(I)$, the attained pointed inner-dual minimum at x, together with a minimum nonzero inner-dual word in a second block, gives a zero-functional pointed nonembedded witness of weight at most $r+1$. This proves necessity as well as sufficiency. Finally, the N targets (j,x) among the mN concatenated coordinates give density $1/m$. □

Proposition 4.2 (Concatenated parameters)**.** *If I has parameters $[m,k,d]_q$ and $O_N\le L^N$ has L-dimension K_N and symbol distance D_N, then $C_N=O_N\circ I$ has length mN, $\mathbb{F}_q$-dimension kK_N, and minimum distance at least dD_N. In particular, if*

$$\frac{K_N}{N}\longrightarrow R>0,\qquad \liminf_N \frac{D_N}{N}\ge\delta>0,$$

then the concatenated family has asymptotic rate kR/m and relative distance at least $d\delta/m$.

Proof. The blockwise inner encoder is injective, so restriction of scalars multiplies the outer L-dimension by $[L:\mathbb{F}_q]=k$. A nonzero outer word has at least D_N nonzero symbols, and each nonzero symbol encodes to an inner word of weight at least d. The finite parameter statements follow; division by mN gives the asymptotic claims. □

Corollary 4.3 (Positive-density MDS coefficient fingerprints)**.** *Let I be an $[m,k]_q$ MDS code with $1\le k<m$, and fix x. Its radius-k support port is the complete k-uniform clutter on the other coordinates, while its radius-k coefficient port reconstructs I, and $z_x(I)=2(k+1)$. Both ports occur at density $1/m$ in every sufficiently long concatenation by an outer family with dual distance tending to infinity.*

Proof. The minimum-port assertions are Theorem 1.1. Every minimum dual support through x has size $k+1$, so $\mu_x(0)=d(I^{\perp})=k+1$ and

$$z_x(I)=2(k+1)>k+1.$$

Apply Theorem 4.1 with $r=k$. The support port depends only on the MDS parameters; the transferred normalized coefficients retain the represented inner code. □

Corollary 4.4 (Positive-density Clebsch repair ports)**.** *Let I be the $[6,3,4]_{11}$ Clebsch hexagon code [15], and fix a coordinate x. The minimal radius-three repair clutter at x is the complete three-uniform hypergraph on the other five coordinates. Hence it has ten minimal repair supports,*

$$\nu_x^{(\leq 3)} = 1, \qquad \tau_x^{(\leq 3)} = 3,$$

and homogeneous helper-survival reliability

$$10s^3(1-s)^2 + 5s^4(1-s) + s^5.$$

Already the radius-three coefficient port at x determines I. Moreover $z_x(I) = 8$, so both the minimum radius-three port and the full radius-five coefficient port occur with density $1/6$ in an asymptotically good fixed-$\mathbb{F}_{11}$ family.

Proof. The dual $I^\perp$ is again an $[6,3,4]_{11}$ MDS code. Every four-set containing x is therefore the support of a dual word, unique up to scalar. The minimal helper sets are consequently all $\binom{5}{3} = 10$ triples on the other coordinates. Their matching and transversal numbers are one and three, and recovery succeeds exactly when at least three of the five helpers survive, giving the displayed polynomial.

At radius three the common-core star of minimum normalized dual words is already a basis of $I^\perp$, by Corollary 4.3. Thus the minimum coefficient port, and hence also the full radius-five port, recovers $I^\perp$ and I.

Finally, the minimum dual-word weight through x is four, so

$$z_x(I) = \mu_x(0) + d(I^\perp) = 4 + 4 = 8.$$

Taking $r = 5$ gives $r + 1 = 6 < 8$. Theorem 4.1 therefore confines every full pointed witness to its inner block and reproduces the coefficient fibers at one of the six coordinate types in every block. Choosing asymptotically good outer codes over $\mathbb{F}_{11^3}$ gives the claimed fixed-$\mathbb{F}_{11}$ family and density. □

Remark 4.5. The support clutter and reliability polynomial in Corollary 4.4 are shared by every $[6,3,4]$ MDS code. The Clebsch-specific information lies in the scalar coefficient fibers: they recover the inner code that is characterized geometrically in [15].

Remark 4.6. Corollary 4.3 applies without a new transfer argument to every represented MDS inner code. This includes the usual arc and projective Reed–Solomon presentations, and the classical MDS code presentations underlying AME constructions, whenever their displayed field and dimension hypotheses hold. Their minimum support ports are generic MDS data; only their normalized coefficient ports distinguish the represented geometric or quantum-code realization.

This is a represented-port theorem: it does not say that every abstract hypergraph is linearly representable. It yields standard scaled asymptotic regions. If $Q = q^k$, random L-linear outer codes of rate R can have relative distances $\delta, \delta^\perp$ whenever

$$H_Q(\delta) < 1 - R, \qquad H_Q(\delta^\perp) < R.$$

For an inner $[m,k,d]_q$ code the concatenated parameters satisfy

$$R_{\text{concat}} \to \frac{k}{m} R, \qquad \delta_{\text{concat}} \geq \frac{d}{m} \delta.$$

Indeed, for a random L-linear outer code the expected numbers of nonzero words and dual words below the two cutoffs have exponential rates $R + H_Q(\delta) - 1$ and $H_Q(\delta^\perp) - R$, respectively, so both vanish simultaneously; this is the random-linear-code form of the Gilbert–Varshamov argument [12, Chapter 17]. When Q is square, algebraic-geometry outer codes similarly give, for $\gamma = 1/A(Q)$ and $\gamma < R < 1 - \gamma$,

$$\delta(O_N) \geq 1 - R - \gamma, \qquad \delta(O_N^\perp) \geq R - \gamma, \qquad \delta_{\text{concat}} \geq \frac{d}{m}(1 - R - \gamma).$$

These are the classical GV and TVZ ingredients. For the completed $[20, 4, 9]_9$ seed, $Q = 6561$ and $\gamma \leq 1/80$, recovering the concrete rate-1/10 family as one instance [19, Chapters 2 and 7]. The stronger self-dual family used for that concrete instance is precisely Stichtenoth's theorem over $\mathbb{F}_{6561}$ [18, Theorem 1.6(ii)].

5 Reliability and bounded EXIT

Let $\mathcal{H}$ be a repair clutter on helper set V. For $v \in V$, define

$$\mathcal{H} - v = \min\{E \in \mathcal{H} : v \notin E\}, \qquad \mathcal{H}/v = \min\{E \setminus \{v\} : E \in \mathcal{H}\}.$$

Theorem 5.1 (Reliability calculus)**.** *For independent helper survival probabilities (s_v),*

$$R_\mathcal{H}(s) = (1 - s_v)R_{\mathcal{H}-v}(s) + s_v R_{\mathcal{H}/v}(s),$$

and

$$\frac{\partial R_\mathcal{H}}{\partial s_v} = R_{\mathcal{H}/v} - R_{\mathcal{H}-v} = \Pr(v \textit{ is pivotal}).$$

Under homogeneous survival s, $R'_\mathcal{H}(s) = \sum_v \Pr_s(v$ is pivotal$)$. If the minimum blocker size is τ and there are b_τ minimum blockers, then for helper failure probability p,

$$1 - R_\mathcal{H}(1 - p) = b_\tau p^\tau + O(p^{\tau+1}).$$

Proof. Write $f_\mathcal{H}(A) = 1$ when A contains an edge of $\mathcal{H}$, and write it as zero otherwise. The reliability is the finite multilinear sum

$$R_\mathcal{H}(s) = \sum_{A \subseteq V} f_\mathcal{H}(A) \prod_{y \in A} s_y \prod_{y \notin A} (1 - s_y).$$

Partition the subsets according to whether they contain v. Erasing v from those that do contain it identifies the two parts with the Boolean systems $\mathcal{H} - v$ and $\mathcal{H}/v$, respectively. This gives

$$R_\mathcal{H} = (1 - s_v)R_{\mathcal{H}-v} + s_v R_{\mathcal{H}/v}.$$

The two conditional reliabilities do not involve s_v, so differentiation gives $R_{\mathcal{H}/v} - R_{\mathcal{H}-v}$. Monotonicity makes this difference the conditional probability that adjoining v changes failure into success, which is precisely pivotality. On the diagonal $s_v = s$, the chain rule sums these coordinate derivatives and gives the Russo–Margulis identity.

Now expose the failed set F. Repair fails exactly when F is a transversal of $\mathcal{H}$, equivalently when it contains a minimal blocker. Thus

$$1 - R_\mathcal{H}(1 - p) = \sum_{\substack{F \subseteq V \\ F \text{ is a transversal}}} p^{|F|}(1 - p)^{|V| - |F|}.$$

There are no summands below degree τ, and the transversals of size τ are exactly the b_τ minimum blockers. Their contribution is $b_\tau p^\tau (1-p)^{|V|-\tau} = b_\tau p^\tau + O(p^{\tau+1})$. Every remaining term contains at least $p^{\tau+1}$; the sum is finite, proving the expansion. □

These are the standard deletion–contraction and pivotality identities for a monotone reliability system [4].

For EXIT conventions, condition the target x unavailable and erase helper v with probability p_v. Put

$$h_x^{(\le r)}(p) = \Pr(f_x^{(\le r)}(A) = 0), \qquad R_x^{(\le r)} = 1 - h_x^{(\le r)}.$$

This is an *extrinsic* failure probability. If the target is itself erased with probability p_x, its residual erasure probability is $p_x h_x^{(\le r)}$. At full radius it is symbol-MAP EXIT; at finite radius it is a bounded-query decoder and carries no capacity claim [1, 13].

Proposition 5.2 (Bounded-EXIT hierarchy)**.** *The failure form of deletion–contraction is*

$$h_{\mathcal{H}}(p) = p_v h_{\mathcal{H}-v}(p) + (1-p_v) h_{\mathcal{H}/v}(p), \qquad \frac{\partial h_{\mathcal{H}}}{\partial p_v} = \Pr(v \textit{ is pivotal}).$$

If $L_x(A)$ is the size of the cheapest repair contained in the realized survivor set, with $L_x = \infty$ when none exists, then

$$\Pr(L_x = r) = h_x^{(\le r-1)} - h_x^{(\le r)}, \qquad \Pr(L_x = \infty) = h_x^{\mathrm{MAP}}.$$

The first identity is for $r \ge 1$.

Proof. Conditioning on whether v is erased gives the displayed recurrence: when it is erased, only repairs avoiding v remain; when it survives, it may be contracted from every repair. Equivalently, substitute $s_y = 1 - p_y$ in the reliability recurrence and take complements. Differentiation reverses both signs, so the derivative is again the pivotal probability.

Let $\mathcal{H}_x^{(\le r)}$ be the full port filtered by edge size at most r. The event $L_x \le r$ is exactly success for this truncated port. Hence

$$\{L_x = r\} = \{L_x > r-1\} \setminus \{L_x > r\},$$

and taking probabilities gives the difference of failure curves. At full radius, the complementary event is that no repair exists, which is $L_x = \infty$. □

Minimal transversals of $\mathcal{H}_x^{(\le r)}$ are exact target-specific, one-shot bounded-repair failure certificates. They are not automatically the representation-dependent stopping sets of a global iterative Tanner decoder.

For a length-N code define the radius-r locality deficit

$$L_r(x) = \int_0^1 \left(h_x^{(\le r)}(p) - h_x^{\mathrm{MAP}}(p) \right) dp.$$

If Δa_k counts size-k survivor sets repaired by full MAP but not at radius r, beta integration gives

$$L_r(x) = \sum_{k=0}^{N-1} \frac{\Delta a_k}{N\binom{N-1}{k}}.$$

With entropy normalized in $\log_q$ units, the unrecoverable-symbol indicator is the normalized conditional entropy on the q-ary erasure channel. Under this convention the symbol-MAP areas sum to the code dimension K [1], so

$$\sum_x \int_0^1 h_x^{(\le r)}(p)\, dp = K + \sum_x L_r(x).$$

6 Pointed Tutte structure

Let M be the column matroid of a projective-system code on $E = V \sqcup \{x\}$, with x neither a loop nor a coloop. Put

$$\epsilon_x(A) = r_M(A \cup \{x\}) - r_M(A) \in \{0, 1\}.$$

The set A repairs x exactly when $\epsilon_x(A) = 0$.

Theorem 6.1 (Pointed perspective polynomial)**.** *Let $D = M\backslash x$ and $Q = M/x$. The Las Vergnas polynomial of the elementary perspective $D \to Q$ is*

$$T_{D\to Q}(X, Y, Z) = \sum_{A\subseteq V} (X - 1)^{r(M)-r_M(A\cup\{x\})}(Y - 1)^{|A|-r_M(A)}Z^{\epsilon_x(A)}.$$

If

$$S_x(u) = \sum_{\epsilon_x(A)=0} u^{|A|},$$

then, for $n = |V|$,

$$R_{(M,x)}(s) = \sum_{\epsilon_x(A)=0} s^{|A|}(1 - s)^{n-|A|}.$$

Consequently, when $s \neq 1$,

$$R_{(M,x)}(s) = (1 - s)^n S_x\left(\frac{s}{1 - s}\right).$$

Writing $U_K(u, y) = \sum_A u^{|A|}y^{r_K(A)}$, one also has

$$S_x(u) = \frac{\partial}{\partial y}\big(U_D(u, y) - U_Q(u, y)\big)\Big|_{y=1}.$$

Proof. Contraction gives $r_D(A) = r_M(A)$ and $r_Q(A) = r_M(A \cup \{x\}) - 1$, so $r_D(A) - r_Q(A) = 1 - \epsilon_x(A)$. Substitution in the rank-one perspective expansion gives the displayed polynomial term by term. The Z^0 terms are exactly the successful sets; more explicitly, for $u \neq 0$,

$$S_x(u) = u^{r(M)}[Z^0]T_{D\to Q}(1 + u^{-1}, 1 + u, Z),$$

because a term with exponents $a, b, 0$ has $|A| = r(M) - a + b$. Multiplying the coefficient of $u^{|A|}$ by $s^{|A|}(1 - s)^{n-|A|}$ and summing gives the reliability formula. Finally,

$$\partial_y(y^{r_D(A)} - y^{r_Q(A)})\Big|_{y=1} = r_D(A) - r_Q(A),$$

which is one when $\epsilon_x(A) = 0$ and zero when $\epsilon_x(A) = 1$. Summing the differentiated terms proves the last identity. □

This identification is the singleton specialization of standard set-pointed Tutte theory [21, equation (3.1)]; it is not a new polynomial.

Corollary 6.2 (Pointed repair–failure duality)**.** *Pointed duality gives*

$$R_{(M,x)}(s) + R_{(M^*,x)}(1 - s) = 1,$$

and the minimal failure blockers of (M, x) are exactly the minimal repair sets of (M^*, x). For a represented matroid they are the sets $Q' \setminus \{x\}$, where Q' ranges over cocircuits containing x; equivalently, Q' is an inclusion-minimal row-code support through x. In particular,

$$\tau_{\text{full}}(x) = d_x(C) - 1, \qquad d_x(C) = \min\{\text{wt}(c) : c \in C,\ c_x \neq 0\}.$$

Proof. For $F \subseteq V$, put $A = V \setminus F$. The dual rank formula, applied once to F and once to $F \cup \{x\}$, gives

$$r_{M^*}(F \cup \{x\}) - r_{M^*}(F) = 1 - \epsilon_x(A).$$

Thus F repairs x in M^* exactly when its complement fails to repair x in M. Complementation preserves the product measure after replacing s by $1-s$, which proves the reliability identity. It also exchanges minimal dual repairs with minimal primal failure blockers. Circuits of M^* are cocircuits of M, so the blockers are $Q' \setminus \{x\}$ for cocircuits Q' through x. In a represented matroid, cocircuits are the inclusion-minimal nonzero row-code supports. Taking the smallest support through x proves the distance formula. □

Proposition 6.3 (The radius filtration is not a pointed-Tutte specialization)**.** *The complete pointed polynomial does not determine the bounded-radius reliability filtration, even among rank-four systems represented over* $\mathbb{F}_7$.

Proof. The mechanism is sparse paving. In a rank-r sparse paving matroid, every set of size below r is independent, every set of size above r has full rank, and an r-set has rank $r-1$ exactly when it is a circuit-hyperplane. For a distinguished element x, the pointed subset profile therefore depends only on the ground-set size, the rank, the number of circuit-hyperplanes through x, and the number avoiding x. It does not record how the circuit-hyperplanes through x meet after x is removed. The radius-$r-1$ reliability does record those intersections, because it is the probability of the union of their containment events.

Distinguish column 0 in each of the following matrices; columns $1, \ldots, 6$ are the helpers:

$$G_{\mathrm{dis}} = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 1 & 1 \\ 0 & 3 & 1 & 4 & 0 & 6 & 1 \\ 0 & 2 & 3 & 2 & 1 & 5 & 0 \\ 0 & 1 & 4 & 1 & 2 & 4 & 3 \end{pmatrix}, \qquad G_{\mathrm{ov}} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 5 & 4 & 0 & 6 \\ 0 & 4 & 2 & 0 & 2 & 1 & 2 \\ 0 & 4 & 1 & 6 & 4 & 5 & 4 \end{pmatrix}.$$

All entries and determinants below are in $\mathbb{F}_7$. Every three-column set is independent. The dependent four-column sets are, respectively,

$$\{0145, 0236, 2345\}, \qquad \{0123, 0256, 3456\}.$$

Here, for example, 0145 denotes $\{0, 1, 4, 5\}$. For a compact complete minor check, order the four-subsets lexicographically as

$$\begin{gathered} 0123, 0124, 0125, 0126, 0134, 0135, 0136, 0145, 0146, 0156, \\ 0234, 0235, 0236, 0245, 0246, 0256, 0345, 0346, 0356, 0456, \\ 1234, 1235, 1236, 1245, 1246, 1256, 1345, 1346, 1356, 1456, \\ 2345, 2346, 2356, 2456, 3456. \end{gathered}$$

The corresponding determinants are

$$\begin{aligned} G_{\mathrm{dis}} : {} & 5, 3, 3, 5, 4, 4, 1, 0, 5, 5, 2, 5, 0, 6, 5, 2, 1, 1, 6, 4, \\ & 1, 6, 6, 3, 4, 6, 4, 6, 5, 2, 0, 1, 6, 3, 4, \\ G_{\mathrm{ov}} : {} & 0, 2, 5, 1, 2, 5, 1, 5, 2, 6, 1, 6, 4, 4, 5, 0, 5, 4, 4, 5, \\ & 1, 6, 4, 2, 2, 4, 3, 1, 1, 3, 6, 2, 2, 6, 0. \end{aligned}$$

Each triple extends to a displayed nonzero minor, and each set of at least five columns contains one. Hence this list determines every subset rank.

Both matroids are rank-four sparse paving matroids with three circuit-hyperplanes, two through the distinguished column and one avoiding it. The preceding sparse-paving count therefore gives, for both systems,

$$T(X,Y,Z) = (X-1)^3 Z + 6(X-1)^2 Z + 15(X-1)Z + 2(X-1) \\ + 18Z + (Y-1)Z + 14 + 6(Y-1) + (Y-1)^2.$$

Equivalently, their complete successful-set enumerators are both

$$2u^3 + 14u^4 + 6u^5 + u^6.$$

At radius three, however, the minimal repairs are

$$\{145, 236\} \quad \text{for } G_{\text{dis}}, \qquad \{123, 256\} \quad \text{for } G_{\text{ov}}.$$

The first pair is disjoint and the second pair meets in helper 2. Inclusion–exclusion under homogeneous survival consequently gives

$$R_{\text{dis}}^{(\leq 3)}(s) = 2s^3 - s^6, \qquad R_{\text{ov}}^{(\leq 3)}(s) = 2s^3 - s^5.$$

The pointed polynomials agree while the filtered curves do not. □

The retained field-nine harmonic computation gives a geometric instance of the same boundary: six-element circuits in a rank-five code add five-helper repairs beyond the classified radius-four port. It illustrates the effect but is not evidence for Proposition 6.3.

7 Geometric flagships

7.1 The twisted-cubic–axis family

Let $q = 3^h$ and work in $\mathrm{PG}(3,q)$. Put

$$T_q = \{C(t) = (1 : t : t^2 : t^3) : t \in \mathbb{F}_q\},$$

$$L_q = \{A(u) = (0 : 1 : u : 0) : u \in \mathbb{F}_q\} \cup \{A_\infty = (0 : 0 : 1 : 0)\}.$$

The line L_q is the characteristic-three axis of the osculating-plane pencil [5]; all statements below follow directly from the displayed coordinates.

Theorem 7.1 (Cubic–axis port)**.** *The row code on $T_q \sqcup L_q$ is $[2q+1, 4, q-1]_q$. Its minimal circuits of size at most four are the triples on L_q and the sets*

$$\{C(s), C(t), C(u), (0 : e_1 : e_2 : 0)\}$$

for distinct s, t, u, where $e_1 = s + t + u$ and $e_2 = st + su + tu$. Every cubic coordinate has

$$(\nu, \tau) = \left(\frac{q-1}{2}, q-2\right).$$

If $Z_3(q)$ is the maximum size of a three-term-zero-sum-free subset of $(\mathbb{F}_q, +)$, then

$$(\nu, \tau)_{A_\infty} = \left(\frac{5q-3}{6}, 2q-1-Z_3(q)\right),$$

$$(\nu,\tau)_{A(a)} = \left(\frac{5q-9}{6}, 2q-2-Z_3(q)\right).$$

In particular every coordinate satisfies $\tau > \nu$ for $q \geq 9$.

After adjoining $C(\infty) = (0:0:0:1)$, the completed row code is $[2q+2, 4, q]_q$. Radius four is its full minimal port, and its rows are uniform on the two projective blocks:

$$\left(\frac{q-1}{2}, q-1\right) \text{ on the cubic}, \qquad \left(\frac{5q-3}{6}, 2q-3\right) \text{ on the axis.}$$

Proof. A plane containing L_q contains at most one cubic point by Frobenius injectivity; a plane not containing it meets the line once and the cubic at most three times. A plane through L_q and one cubic point has $q+2$ columns, so the largest section has size $q+2$; the length $2q+1$ and rank four therefore give distance $q-1$.

Write $\Delta(s,t,u) = (s-t)(s-u)(t-u)$. Direct expansion gives, up to the fixed ordering sign,

$$\det\big(C(s), C(t), C(u), (0:a:b:0)\big) = \Delta(s,t,u)(ae_2 - be_1).$$

Thus three distinct cubic points meet the axis in the unique point $(0:e_1:e_2:0)$. If $e_1 = 0$, then in characteristic three $e_2 = -(s-t)^2 \neq 0$, so this point is A_∞. Four cubic columns have nonzero Vandermonde determinant. Two distinct cubic and two distinct axis columns also have nonzero determinant, while any set containing three axis points already contains an axis triple. These calculations prove both the displayed circuit list and its completeness through size four.

We record the elementary hypergraph calculation used for the exact rows. Let $\mathcal{Z}$ be the triples of $\mathbb{F}_q$ with sum zero. The parallel class $\{a, a+1, a-1\}$ partitions $\mathbb{F}_q$, so $\nu(\mathcal{Z}) = q/3$; after deleting all triples through zero, deleting the one affected parallel-class member gives $\nu = q/3 - 1$. A set avoids $\mathcal{Z}$ exactly when it is three-term-zero-sum-free, hence $\tau(\mathcal{Z}) = q - Z_3(q)$. Translation preserves $\mathcal{Z}$ and lets a maximum avoiding set omit zero; adjoining the now-isolated zero shows that the deleted hypergraph has transversal number $q - 1 - Z_3(q)$.

At an axis target, the minimal clutter is the disjoint union of the complete graph on the other q axis points and a three-cubic component. At A_∞ that component is $\mathcal{Z}$. At $A(a)$, the bijection

$$s \longmapsto (s+a)^{-1}$$

identifies it with the zero-sum triples avoiding zero: expanding the numerator of the sum of the three inverses gives exactly $e_2 - ae_1 = 0$. Matching and transversal numbers add across the two disjoint helper blocks. Since K_q contributes $((q-1)/2, q-1)$, the two stated axis rows follow.

Fix now a cubic target $C(x)$. Relabel every other cubic parameter by $u = (s-x)^{-1} \in \mathbb{F}_q^*$. The repair using parameters s, t has axis completion

$$\chi_x(u+v), \qquad \chi_x(w) = \begin{cases} A_\infty, & w = 0, \\ A(-x + w^{-1}), & w \neq 0, \end{cases}$$

and χ_x is injective. If g generates $\mathbb{F}_q^*$, pairing g^{2i} with g^{2i+1} uses every nonzero parameter once and gives distinct colors $g^{2i}(1+g)$. This is a matching of size $(q-1)/2$, and no matching can be larger because every repair uses two cubic helpers. For the transversal, all but one cubic helper give a cover of size $q-2$. Conversely, if a cover selects t cubic helpers and leaves a nonempty set U unselected, fix $u_0 \in U$. For every $v \in U \setminus \{u_0\}$, the distinct axis color $\chi_x(u_0 + v)$ must be selected. Hence the cover has at least $t + (|U| - 1) = q - 2$ vertices. This proves the cubic row without an unstated complement lemma. The bound $Z_3(q) \leq q - 2$, witnessed by the zero-sum triple $\{0, 1, -1\}$, gives $\tau > \nu$ for $q \geq 9$.

For the completion, projective shifted inversion $s \mapsto (s+a)^{-1}$, with its pole sent to infinity and infinity to zero, is induced by an invertible ambient map; it is transitive on each of the cubic and axis blocks. It therefore suffices to use targets $C(\infty)$ and A_∞. At $C(\infty)$ the radius-three repairs are $\{C(s), C(t), A(s+t)\}$. The same pairing and covering argument gives $((q-1)/2, q-1)$. At A_∞, the radius-three matching already has $(q-1)/2 + q/3 = (5q-3)/6$ edges. In any matching of minimal radius-four repairs, if an edge uses c cubic and a axis helpers, independence of the helper columns and the circuit classification give $2c + 3a \geq 6$. Summing over the disjoint edges and using the $q+1$ cubic and q available axis coordinates yields $6\nu \leq 2(q+1) + 3q = 5q + 2$, hence the same upper bound $(5q-3)/6$.

Finally, a helper set contains no full repair exactly when it lies in a hyperplane section avoiding the target. For $C(\infty)$, the largest such section has $q+2$ points (the plane $X_3 = 0$ attains it), giving $\tau = (2q+1) - (q+2) = q - 1$. A plane avoiding A_∞ meets the axis once and the cubic at most three times, and an explicit four-point section attains four; hence $\tau = (2q+1) - 4 = 2q - 3$. Rank four makes every minimal circuit have at most five points, so radius four exhausts the full minimal port. $\square$

This is the inner code in the strict weighted-transfer example above. The exact $q = 9$ rows are recorded in Appendix A.1.

7.2 The quartic–nucleus harmonic family

Let

$$V(t) = (1, t, t^2, t^3, t^4), \qquad V(\infty) = e_4, \qquad N = e_2,$$

and let $S_q = \{V(t) : t \in \mathbb{P}^1(\mathbb{F}_q)\} \cup \{N\}$. The classical normal-rational-curve nucleus formula identifies N as the hyperplane nucleus in characteristic three [6, Theorem 1].

Theorem 7.2 (Quartic–nucleus port)**.** *For $q = 3^h \geq 9$, the row code Q_q on S_q has parameters*

$$[q+2, 5, q-3]_q, \qquad d(Q_q^\perp) = 5.$$

Its minimal circuits of size at most five are exactly

$$\{N\} \cup B,$$

where B is a harmonic quadruple of $\mathbb{P}^1(\mathbb{F}_q)$. These quadruples form a Steiner system $S(3, 4, q+1)$, the characteristic-three four-set orbit in the standard $\mathrm{PGL}(2, q)$ design construction [20, Section 2]. Hence the radius-four repairs are all B at the nucleus and all $\{N\} \cup (B \setminus \{x\})$ at a curve target $x \in B$. The nucleus has

$$\frac{(q+1)q(q-1)}{24}$$

radius-four repairs, and every curve target has $q(q-1)/6$. Every curve target has $(\nu, \tau) = (1, 1)$.

Proof. For distinct finite a, b, c, d, the hyperplane through their quartic columns contains N exactly when $e_2(a, b, c, d) = 0$; with ∞ present the condition is $a + b + c = 0$. Given three finite points, if $e_1 = a + b + c \neq 0$, the unique completion is $d = -e_2/e_1$; if $e_1 = 0$, the unique completion is ∞. In the first case, substituting $d = a$, for example, changes the harmonic equation to $(a-b)(a-c) = 0$, and similarly for b and c; hence the completion is distinct. In the second case, writing $c = -a - b$ gives $e_2(a, b, c) = -(a-b)^2 \neq 0$, so no finite completion exists. A triple $\{\infty, a, b\}$ has the unique completion $-a-b$, distinct from a, b in characteristic three. This proves the Steiner property. Since

$\{\infty, 0, 1, -1\}$ is one block and $\mathrm{PGL}(2, q)$ is 3-transitive, unique completion also identifies the blocks with its projective orbit.

Any five curve points are independent by the extended Vandermonde determinant. Expanding the determinant of $N, V(a), V(b), V(c), V(d)$ along the nucleus row gives

$$\Delta(a, b, c, d)e_2(a, b, c, d),$$

and with infinity present it gives $\Delta(a, b, c)e_1(a, b, c)$. The Vandermonde factor is nonzero for distinct parameters. Thus $\{N\} \cup B$ is dependent exactly for a harmonic block. A complementary minor shows that N with any three distinct curve points is independent; in the zero-sum finite branch its determinant reduces to $-\Delta(a, b, c)e_2(a, b, c)$, which is nonzero by the calculation above. Deleting any point therefore restores independence. These are all circuits of size at most five, so the dual distance is five.

A nonzero hyperplane restricts to a nonzero polynomial of degree at most four on the normal rational curve, counting infinity as a root when the quartic coefficient vanishes. It therefore contains at most four curve points. If it also contains N, equality occurs exactly on a harmonic block, so the largest section of S_q has five points. Hence the primal distance is $(q + 2) - 5 = q - 3$. Finally, the Steiner count is $\binom{q+1}{3}/\binom{4}{3}$, and the number of blocks through a fixed curve point is $\binom{q}{2}/\binom{3}{2}$. The circuit–repair correspondence gives the stated ports and counts. □

The inner-dual half of the coarse transfer gate is automatic at radius four because $5 < 2d(Q_q^\perp) = 10$. Theorem 4.1 therefore places the entire harmonic port with positive density in asymptotically good fixed-$\mathbb{F}_q$ families.

7.3 Reliability, EXIT, and the geometric contrast

The nucleus repairs are the blocks of the Steiner system $S(3, 4, q + 1)$. At a curve target, every radius-four repair contains the nucleus. Thus the nucleus port is a parallel family of four-helper repairs, whereas the curve port has a compulsory series helper. The reliability and bounded-EXIT theorem applies to both ports without an exact finite profile. Exact field-nine profiles are computational refinements recorded in Appendix A.1; they are not used in the body theorem chain.

Finally, the harmonic small-circuit closure has a deterministic nucleus gate. Let $D(S)$ close curve set S under "three points of a harmonic block add the fourth." Then

$$\mathrm{cl}_4(S) = S \quad \text{if } S \text{ contains no block,} \qquad \mathrm{cl}_4(S) = \{N\} \cup D(S) \quad \text{otherwise,}$$

and $\mathrm{cl}_4(S \cup \{N\}) = \{N\} \cup D(S)$. Indeed, without N no curve point can be repaired, while a complete block $B \subseteq S$ repairs N. Once N is present, the curve-target repairs are exactly N together with three points of a harmonic block, so iteration is precisely harmonic completion. This proves all three identities.

Appendix A.1 gives a field-nine block-free rank-five spanning set and its one-round completion after adjoining N. These witnesses illustrate the nucleus gate but are not inputs to the closure identities, an asymptotic propagation claim, or a threshold theorem.

8 Conclusion

A complete bounded repair port is one pointed object with three compatible views. Its support clutter measures disjoint availability and adversarial tolerance; its coefficient fibers give exact scalar recovery and the correct weighted transfer cost; its Boolean function gives reliability and bounded

EXIT. Exact concatenation transfers the whole object, and the persistent pointed obstruction is necessary and sufficient for eventual confinement in the fixed-inner linear-concatenation construction used here. Below that gate, the construction realizes the represented port at positive density.

The cubic–axis family shows rich overlapping ports and strict weighted transfer beyond a support-only gate. The quartic–nucleus family replaces that overlap pattern by a harmonic Steiner design: nucleus repairs form a parallel block family while curve targets share a compulsory series helper. Pointed-Tutte theory explains the full-radius structure; the radius filtration records when each repair first becomes available.

A sharper cubic blocker-stability statement is not needed for the present results. Sequential composition, general service regions, coefficient optimization, log-concavity, and random harmonic-cascade thresholds lie outside this paper.

A Formal verification and finite evidence

The coefficient-port reconstruction and exact transfer chains are checked in Lean. The import-only gate `RepairPorts.Gates.CompletePorts` audits the paper-facing declarations. Its two transfer terminals are

```
RepairPorts.exactFunctionalStrata,
RepairPorts.exactPointedConfinementAndTransfer.
```

The first gives the exact zero-, singleton-, and multisupport-functional profiles; the second gives the pointed minimum formula and complete-port equality. Both report only `propext`, `Classical.choice`, and `Quot.sound`. The strict field-nine corollary is the conclusion of

```
RepairCodes.projectiveAxisTwistedCubic_strict_weighted_transfer_of_regular_
projective_action.
```

Singer regularity is an explicit hypothesis of that declaration, not a hidden formal axiom.

The completed cubic parameters and rows are independently exposed by

```
FiniteGeom.projectiveAxisTwistedCubic_code_parameters,
RepairCodes.minimalProjectiveAxisTwistedCubicRepair_full_eq_four,
RepairCodes.minimalProjectiveCubicRepair_four_invariants,
RepairCodes.minimalProjectiveAxisRepair_four_invariants.
```

The gate also prints the corresponding full-radius invariant terminals.

The paper's `verification/formal-boundary.json` fixes the formal source commit, the finitegeom base commit, Lean and mathlib revisions, gate-fact hash, 36-module closure, and 60 paper-facing terminals. At source commit `e1b58fd8a79a763012a21fa4d32f8904c8d9eb04`, the immutable area-export planner reports 885232 closure bytes and no prose drift against finitegeom commit `b871c10b4a91200a0913644d39b9f0ce44f655ca`. The companion `verification/README.md` gives the guarded build, axiom-audit, and area-export replay commands. These identifiers distinguish the kernel-checked module closure from the manuscript PDF and from evidence-only finite calculations.

Trace duality, scaled concatenated parameters, exact target density, eventual necessity and sufficiency, support/coefficient transfer, and the MDS fingerprint corollary are audited through

```
RepairPorts.concatenatedRestrictedCode_parameters,
RepairPorts.eventually_pointedConfinement_iff_zeroCost,
RepairPorts.eventually_prescribedPorts,
RepairPorts.eventually_mdsMinimumCoefficientFingerprints.
```

They report only `propext`, `Classical.choice`, and `Quot.sound`. Random-GV or AG/TVZ existence of an outer family is the explicit classical input; no such existence theorem occurs in the Lean dependency closure.

The finite reliability and bounded-EXIT calculus is audited through

```
RepairPorts.portReliability_delete_contract,
RepairPorts.hasDerivAt_portReliability_update,
RepairPorts.hasDerivAt_homogeneous_portReliability,
RepairPorts.erasureFailureProbability_delete_contract,
RepairPorts.noRepairProbability_eq_erasureFailure,
RepairPorts.cheapestRepairRadiusProbability_eq_failure_sub,
RepairPorts.blockerFailurePolynomial_eq_minimum_term_add_remainder.
```

These declarations use finite sums and polynomial algebra only. Exact finite profiles and Poisson approximations remain computational refinements outside the body proof chain.

For the concrete completed field-nine family, the external existence input is Stichtenoth's self-dual TVZ-family theorem over $\mathbb{F}_{6561}$ [18, Theorem 1.6(ii)]. Classical inputs are labeled as such: outer-family existence, concatenated-dual decomposition, Singer regularity, normal-rational-curve nuclei, and the Las Vergnas perspective polynomial.

A.1 Field-nine evidence

For the uncompleted field-nine cubic–axis code on $T_9 \sqcup L_9$, $Z_3(9) = 4$, and the cubic, finite-axis, and axis-infinity coordinate types have exact radius-three (ν, τ) rows

$$(4, 7), \qquad (6, 12), \qquad (7, 13),$$

with multiplicities $9, 9, 1$. These are not rows of the completed code. After adjoining $C(\infty)$, Theorem 7.1 gives the two uniform radius-four/full-port rows

$$(4, 8) \quad \text{on the ten cubic coordinates,} \qquad (7, 15) \quad \text{on the ten axis coordinates.}$$

Corollary 3.2 uses the completed 20-coordinate code. Its generalized-SPC check has coefficients $(1, a, 1, 1, 1)$, where a is a linear Singer multiplier whose translate of the 20 unit-cost projective functional classes is disjoint from the original 20-set. Regularity on 820 classes and $20^2 < 820$ prove that such an a exists. Thus this is an exact conditional-on-Singer existence proof, not a claim that five field elements or a hidden finite multiplier certificate were computed. The functional support distance is five, the weighted and exact nonembedded thresholds are six, and radius-four transfer follows.

For the quartic–nucleus code at $q = 9$, the general formulas give 30 nucleus repairs and 12 repairs at each curve target. The certificate additionally gives nucleus values $(\nu, \tau) = (2, 5)$, exact Bernstein and EXIT rows, and finite propagation witnesses. In particular, a block-free rank-five spanning set completes in one round after adjoining the nucleus. These exact outputs are evidence-only refinements: no theorem in the main body depends on them.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author's direction, the systems assisted throughout with proof exploration, literature research, symbolic and finite computation, code and formal-proof development, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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